IDX G9 GEOMETRY S+ STUDY GUIDE ISSUE S2 FINALS

By Aiden and Julie, Edited by Indy

This guide is a combined guide. Due to the final range including content from before the monthlies, we have included content from the previous issue in this guide for your convenience.

Coordinate Geometry (Points, Lines, and Circles)

<u>Content:</u>

1. **points and lines**
2. **slopes of lines**
3. **finding equations of lines**
4. **circles in the coordinate plane**
5. **proofs using coordinate geometry**
6. **practice questions**

1. Coordinate Geometry Basics

Key Concepts

- **Rectangular (Cartesian) Coordinate System**:
  - **x-axis**: Horizontal number line.
  - **y-axis**: Vertical number line, perpendicular to the x-axis.
  - **Origin**: Point of intersection of the x- and y-axes, (0,0).
  - **Quadrants**: Four regions created by the axes (1,2,3,4).
- **Ordered Pairs**: Represent points as (x, y), where:
  - **Abscissa (x-coordinate)**: Distance from the y-axis.
  - **Ordinate (y-coordinate)**: Distance from the x-axis.

2. Linear Equations

Key Definitions

- A set (S) of ordered pairs (x, y) can be represented by a point in the plane.
- **x-intercept**: Where the graph intersects the x-axis (y = 0).
- **y-intercept**: Where the graph intersects the y-axis (x = 0).

Forms of Linear Equations

1. **General Form**: (Ax + By = C)
2. **Slope-Intercept Form**: (y = mx + k), where:
   - (m): Slope.
   - (k): y-intercept.
3. **Point-Slope Form**: (Y - y1 = m (X - x1)), where (x1, y1) is a point on the line.
4. **Intercept Form**: (x/a+ y/b = 1), where (a) is the x-intercept and (b) is the y-intercept.

Finding Intercepts

- **Example**: For (2x + 3y = 15):
  - **y-intercept**: Set (x = 0): (3y = 15 y = 5). So, y-intercept is (0, 5).
  - **x-intercept**: Set (y = 0): (2x = 15 x = 7.5). So, x-intercept is (7.5, 0).

---

3. Distance and Midpoint Formulas

Formulas

- **Distance Formula**: For points (A (x1, y1)) and (B (x2, y2)): [AB $=\sqrt{(x2-x1)^2+(y2-y1)^2}$.]
- **Midpoint Formula**: For the midpoint (M) of segment (AB): **[M = (x1+x2)/2, (y1+y2)/2]**

5. Slopes of Lines

Slope Formula

- For points (x1, y1) and (x2, y2): **[m = y2-y1/x2-x1]**
- **Horizontal lines**: Slope (m = 0).
- **Vertical lines**: Slope is undefined.
- **Positive slope**: Line rises (as (x) increases, (y) increases).
- **Negative slope**: Line falls (as (x) increases, (y) decreases).

Parallel and Perpendicular Lines

- **Parallel Lines**: Same slope **((m1 = m2)).**
- **Perpendicular Lines**: Slopes are negative reciprocals **(m1* m2 = -1).**

---

6. Circles in the Coordinate Plane

Standard Equation of a Circle

- Center ((h, k)), radius (r): [$(\boldsymbol{x}-\boldsymbol{h})^{2}+(\boldsymbol{x}-\boldsymbol{k})^{2}=\boldsymbol{r}^{2}$]

Tangent to a Circle

- A line is tangent to a circle if it intersects at exactly one point.

7. Proofs Using Coordinate Geometry

Midsegment Theorem (Trapezoid)

- The midsegment of a trapezoid is parallel to the bases and its length is the average of the lengths of the bases.

8.

Find an equation in the **most convenient form** of the line described.

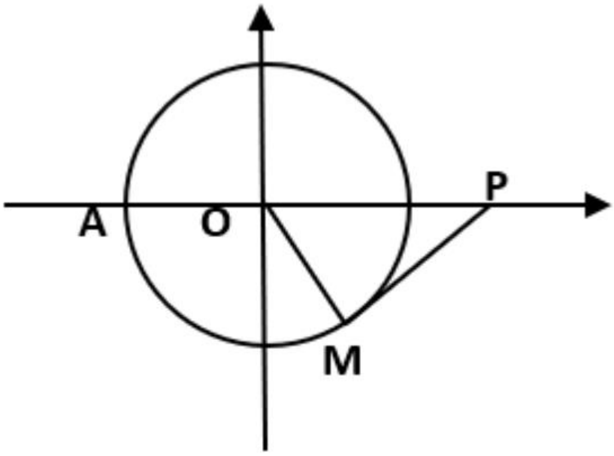
1. The line with x-intercept -3 and y-intercept 8.
2. The line with x-intercept -2 and through $(3,7)$.
3. The line pass through (8,5) and perpendicular to y = 4.
4. The line pass through (9,2) and origin.

5. A quadrilateral with vertices$(-4,4)$, $(-6,0)$, $(-4,-4)$, $(-2,0)$. **Name** the type of quadrilateral and **find its area**. (No proof is required.)

6. $ABCD$ is a square with vertices $A(0,0)$ and $B(5,3)$. Find the coordinates of the point C and D.

7. $\triangle ABC$ has vertices $A(-2,-1)$, $B(0,7)$,and $C(8,3)$.
   1) Find an equation of the median from A.
   2) Find an equation of the altitude from C.
   3) Find the circumcenter Q of $\triangle$ABC

7. Radius of circle O is 5. $OP = 10$. $\overline{PM}$ is tangent to circle O at M.
   a. Find the equation of this circle.
   b. Calculate the measure of $\angle OPM$ and the coordinates of M.



Answer key:

1. x/-3 + y/8 =1
2. y-7=7/5 (x-3)
3. x=8
4. y=2x/9
5. Rhombus; 16
6. D(-3,5) C(5-3,5+3) or D(3,-5) C(8,2)
7. 1) y=x+1

   2) y=-x/4+7

   3) (23/9, 19/9)

7. a. x 平方+y 平方=25

   b. 30 degrees; M(2.5,-2.5 更号 3)

Contents:

1. 1-2 Drawings, Nets, and other Models

1-New Term

- ***Isometric Drawing***

An isometric drawing is a 2D representation of a 3D object where all three dimensions (length, width, and height) are drawn at the same scale and at 30-degree angles from the horizontal.

Key features for drawing:

30° Angles – Horizontal edges are drawn at **30°** from the baseline (vertical lines remain straight).

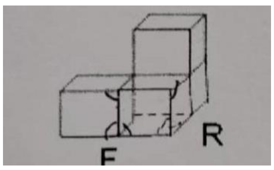
Equal Scaling – All three axes (X, Y, Z) use the **same scale** (no foreshortening).

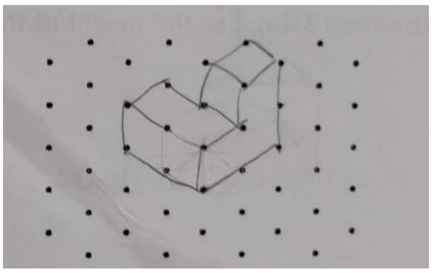
No Perspective – Parallel lines **do not converge** (unlike perspective drawings).

3D Representation – Shows **width, height, and depth** in a single view.

No Hidden Lines – Normally, only visible edges are shown (unless necessary for clarity).

E.g. ↓ 's Isometric Drawing → → →





- ***Orthographic Drawing***

An orthographic drawing is a 2D technical representation of a 3D object, showing multiple flat views (e.g., front, top, side) at right angles to each other.

Key features for drawing:

Multiple 2D Views – Shows **front, top, and right views** (sometimes more).

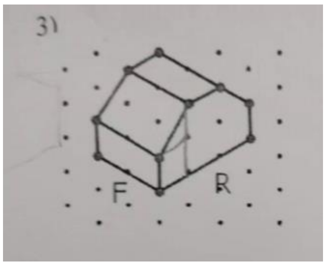
90° Alignment – Views are **perfectly aligned** (top view directly above front view, side view to the side).

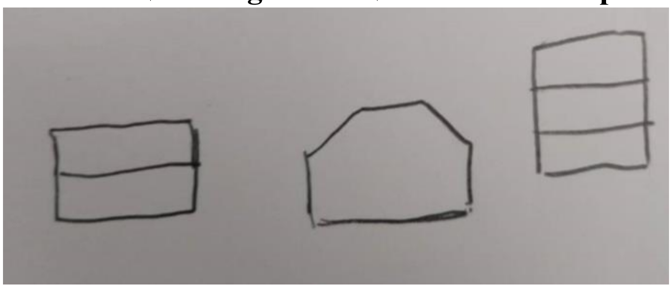
True Dimensions – Each view is **drawn to scale** without distortion.

Hidden Lines – Uses **dashed lines** to show obscured edges.

No Depth Illusion – Pure 2D representation (unlike isometric or perspective).

E.g. **Front view ↓** **Right view ↓** **Top view ↓**





- ***<u>Foundation Drawing</u>***

Foundation Drawing is a 2D technical representation of a 3D object, usually showing the top view of it. Each small figures have a number on it, represent its height.

Key Features for drawing:

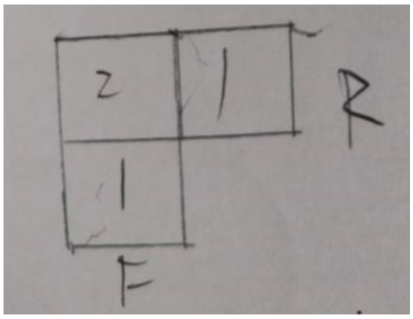
Top 2D View - Just show the **top view,** but also show the **direction** of it**.**

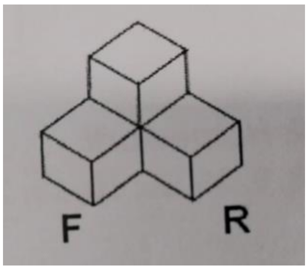
90° Alignment – Views are **perfectly aligned** (directly above front view).

True Dimensions – Each view is **drawn to scale** without distortion.

Entire arrangement - The figure should be divided into several **regular cubes with edge 1 unit**.

Counting necessity - Each small cubes should have a **number** of it to represent its **height**(how many cubes are used to stack it).

e.g. ↓ 's Foundation Drawing → → → 



• ***<u>Net</u>***

A net is a 2D pattern that can be folded along its edges to form a 3D geometric solid (e.g., cube, pyramid, prism).

Key Features for drawing:

Complete Face Representation - Includes **every face** of the 3D shape (no missing or overlapping parts).

Edge Alignment - Edges that will join in 3D must match in length and position in the 2D net.

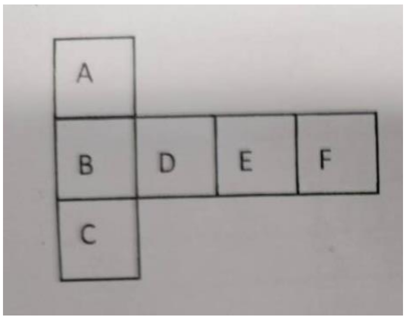
No Gaps or Overlaps - faces must connect **edge-to-edge** without extra space or unintended overlaps.

Foldable Design - Must be physically possible to fold into the intended 3D shape (e.g., valid cube nets have 11 distinct configurations).

Scale Accuracy - All faces must be **proportionally sized** to match the 3D solid.

e.g.

The net of a cube ↓



2. __11-1 Space Figures and Cross Sections__

1-New Term

- ***Polyhedron***

A Polyhedron is a three-dimensional shape.

- ***Face***

Each face is a face of the polyhedron.

- ***Edge***

An edge is a segment that is formed by the intersection of two faces.

- ***Vertex***

A vertex is a point where three or more edges meet.

- ***Cross Section***

A cross section is the shape formed of a solid and a plane. You can think of a cross section as a very thin slice of the solid.

2-New Formula

- ***Euler's Formula***

The numbers of faces(F), vertices(V), and edges(E) of a polyhedron are related by F+V=E+2. In two dimensions, Euler's formula reduces to F+V=E+1 where F is the number of regions formed by V vertices linked by E segments.

3. __11-2 Surface Areas of Prisms and Cylinders__

1-New terms

- ***Prism***

A Prism is a polyhedron with at least two congruent parallel faces, called lateral faces. Other faces are two opposite faces called bases. You name a prism by the shape of its base.

- ***Lateral Area***

The lateral area of a prism is the sum of the areas of the lateral faces.

- ***Surface Area***

The surface area is the sum of the lateral area and the area of the two bases.

- ***Cylinder***

A cylinder has two congruent parallel bases, which are circle. Unrolling the curved face results in the rectangle . The area of this resulting rectangle is the product of the circumference of base and height. The surface area is the sum of the lateral area and the area of the two congruent bases.

- ***Altitude***

An altitude of a prism(or a cylinder) is a perpendicular segment that joins the planes of the bases.

- ***Height***

The height (h) of the prism(or a cylinder) is the length of an altitude.

A prism(or a cylinder) may either be right or oblique.

2-New Theorem

Theorem 11-1(Lateral and Surface Areas of a Prism) **The lateral area of a right prism is the product of the circumference of the base and the height. L.A.=Ch.The surface area of a right prism is the sum of the area of bases and the lateral areas. S.A.=Ch+2B**

Theorem 11-2(Lateral and Surface Areas of a Cylinder) **The lateral are of a right cylinder is the product of the circumference of the base and the height. L.A.=Ch. The surface area of a right cylinder is the sum of the area of bases and the lateral areas. S.A.=Ch+2B, or S.A.=Ch+2 π r^2**

4. 11-3 Surface Areas of Pyramids and Cones

1-New terms

- ***Pyramid***

A pyramid is a polyhedron in which one face (the base) can be any polygon and other faces(the lateral faces) are the triangular sides that meet at a common vertex (vertex of the pyramid). We name a pyramid by the shape of its base.

- ***Altitude***

The altitude is the perpendicular segment from the vertex of the pyramid to the plane of the base. The length of the altitude is the height h.

- ***Slant height***

The slant height (l) is the length of the altitude of a lateral face of a pyramid.

- ***Regular pyramid***

A regular pyramid is a pyramid whose base is a regular polygon and whose lateral faces are congruent isosceles triangles.

- ***Cones***

The base of a cone is a circle.

- ***Altitude***

In a right cone, the altitude is a perpendicular segment from the vertex to the center of the base. The height(h) is the length of the altitude,

- ***Slant height***

The slant height (l) is the distance from the vertex to a point on the circumference of the base.

2-New Theorem

Theorem 11-3(Lateral and Surface Areas of a Regular Pyramid) **L.A.=Cl, S.A.=L.A.+B**

Theorem 11-4(Lateral and Surface Areas of a Cone) **L.A.=$\pi r l$, S.A.= L.A.+πr^2**

5. 11-4 Volumes of Prisms and Cylinders

1-New terms

- ***Volume***

Volume is the measure of the amount of 3-dimensional space occupied by a solid object, expressed in cubic units (e.g., cm^3, m^3, in^3). It quantifies the capacity or "how much a container can hold."

2-New Theorem

Theorem 11-5(Cavalierl's Principle) **If two space figures have the same height and the same cross-sectional area at every level, then they have the same volume.**

Theorem 11-6(Volume of a Prism) **$V=Bh$**

Theorem 11-7(Volume of a Cylinder) **$V=Bh=\pi r^2 h$**

6. <u>11-5 Volumes of Pyramids and Cones</u>

1-New Theorem

Theorem 11-8(Volume of a Pyramid) **$V=\frac{1}{3}Bh$**

Theorem 11-9(Volume of a Cone) **$V=\frac{1}{3}Bh$**

7. <u>11-6 Surface Areas and Volumes of Spheres</u>

1-New terms

- ***<u>sphere</u>***

A sphere is the set of all points in three-dimensional space equidistant from a fixed point called the center.

- ***<u>radius</u>***

A radius is a segment that has one endpoint at the center and the other endpoint on the sphere.

- ***<u>diameter</u>***

A diameter is a segment passing through the center with endpoints on the sphere.

- ***<u>great circle</u>***

A great circle is the largest possible circle of the sphere that contains the center of the sphere. A great circle divides a sphere into two hemispheres.

2-New Theorem

Theorem 11-10(Surface Area of a Sphere) **S.A.=**$4\pi r^2 = \pi d^2$

Theorem 11-11(Volume of a Sphere) **V=**$\frac{4}{3}\pi r^3$

8. 11-7 Areas and Volumes of Similar Solids

1-New Theorem

Theorem 11-12(Areas and Volumes of Similar Solids) If the S.R. of two similar solids is a:b, then $\frac{SA_1}{SA_2} = \frac{a^2}{b^2}, \frac{V_1}{V_2} = \frac{a^3}{b^3}$